Financial Decision Making and the Law of One Price

International Corporate Finance – 1

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Based on:
Berk – DeMarzo: Corporate Finance. 6th edition, Pearson – 3rd and 5th chapter

Valuation principle

The value of an asset is determined by its competitive market price.

Competitive market: goods can be bought and sold at the same price.

Price = cash value of the good.

Value **does not** depend on the **views** and **preferences** of the decision maker.

Valuation principle – Example 3.1

Competitive Market Prices Determine Value

Problem

You have just won a radio contest and are disappointed to find out that the prize is **four tickets** to the Def Leppard reunion tour (**face value $40 each**). Not being a fan of 1980s power rock, you have no intention of going to the show. However, there is a second choice: **two tickets** to your favorite band's sold-out show **(face value $45 each).** You notice that on eBay, tickets to the Def Leppard show are being **bought and sold for $30 a piece** and tickets to your favorite band's show are being **bought and sold at $50 each**. Which prize should you choose?

Valuation principle – Example 3.1 – solution

Competitive Market Prices Determine Value

Solution

Competitive market prices, **not your personal preferences** (**nor the face value** of the tickets), are relevant here:

- Four Def Leppard tickets at $30 a piece = $120 market value
- Two of your favorite band's tickets at $ 50 a piece = $100 market value

Instead of taking the tickets to your favorite band, you should accept the Def Leppard tickets, sell them on eBay, and use the proceeds to buy two tickets to your favorite band's show. You'll even have $ 20 left over to buy a T-shirt.

Benefits – costs

Financial manager’s task is to **make decisions** on behalf of the investors that **increase the value of the firm**.

If an investment’s value of **benefits** > value of **cost** → increase market value of the firm.

Important: cost and benefits should be in common terms to be able to compare them!!!

1. ***same time → convert with interest rate → time value of money***
2. ***same currency → convert with exchange rate***
3. ***same risk → return-risk relationship***

Returns…

EAR – effective annual return → WE USE IT FOR PRICING!

APR - annual percentage return → interest rate per compounding period:

APR/(k periods/year) & APR is USED TO DEFINE THE CASH FLOW

$(1 + EAR) = (1 + APR/k)^k$ → if countinously compounding: $(1+EAR) = e^{APR}$

$(1+ r_{real}) = (1 + r_{nominal})/(1 + \text{inflation})$ → appr. $= r_{nominal}$ – inflation rate

$r_{after\ tax} = r \times (1 - \text{tax rate})$

Annual Percentage Rates

Effective Annual Rates for a 6% APR with Different Compounding Periods

Compounding Interval	Effective Annual Rate
Annual	$\left(1+\frac{0.06}{1}\right)^{1}-1=6\%$
Semiannual	$\left(1+\frac{0.06}{2}\right)^{2}-1=6.09\%$
Monthly	$\left(1+\frac{0.06}{12}\right)^{12}-1=6.1678\%$
Daily	$\left(1+\frac{0.06}{365}\right)^{365}-1=6.1831\%$

- A 6% APR with continuous compounding results in an EAR of approximately 6.1837%.=exp(0.06)-1

Time value of money

t	C	IRF	r	y
0	100			
1	80	0.8	-20.00%	-22.31%
2	100	1.25	25.00%	22.31%

Future value → (1+r) Interest Rate Factor (IRF) → *it gives the value of today's dollar in a year.*

$$C_0(1+r_t)^t = C_t = C_0 e^{y_t t}$$

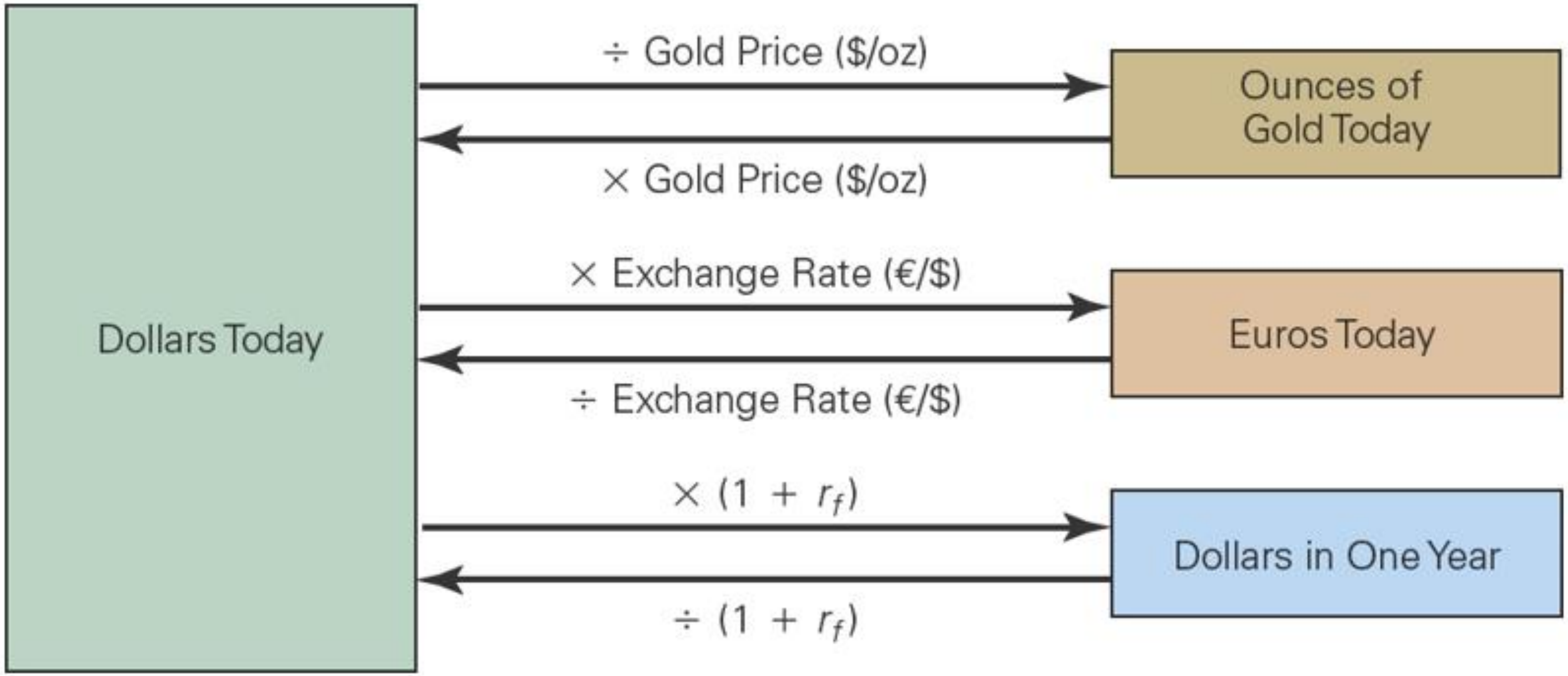
Present value → 1 / (1+r) Discount Factor (DF) → *it gives today's value of one dollar in a year.*

$$C_0 = \frac{C_t}{(1+r_t)^t} = \frac{C_t}{e^{y_t t}}$$

y → logreturn *(r)*

$$(1+r) = e^{\ln(1+r)} = e^y$$

Conversions



Net Present Value

NPV = PV(Benefits) – PV(Costs) = PV(all project cash flows)
NPV > 0 accept
NPV < 0 reject
When making an investment decision, take the alternative with the highest NPV.
Choosing this alternative is equivalent to receiving its NPV in cash today.

Regardless of our **preferences** for **cash today versus cash in the future**, we should **always maximize NPV first**.
We can then borrow or lend to shift cash flows through time and find our most preferred pattern of cash flows.

Value additivity: PV(A) + PV(B) = PV(A+B)

Arbitrage and the Law of One Price

Arbitrage: The practice of buying and selling equivalent goods in different markets to take advantage of a price – **make a profit** – difference **without** taking any **risk**.

Normal market: a competitive market in which there is no arbitrage opportunities.

Law of One Price: If equivalent investment opportunities trade simultaneously in different competitive markets, then they must trade for the same **same price in both markets**.

↓

Used in derivative pricing

Arbitrage example

Assume a security promises a risk-free payment of $1000 in one year. If the risk-free interest rate is 5%, what can we conclude about the price of this bond in a normal market?

Price = 1000 / (1+ 5%) = 952.38

What if the price is different on the market, what would you do? Let’s say it is only 940 or it is 960…

Arbitrage example – solution

1. step: identify the assets in the arbitrage.
2. step: identify which asset is too cheap, and which one is too expensive.
3. step: **buy the asset that is cheap, and sell the one that is expensive!**
4. remark: you should buy or sell every asset which are mispriced **compared** to the other!
5. remark: you **NEVER** use your own money!

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	Today($)	**In One year($)**
Buy the bond	−940.00	+1000.00
Borrow from the bank	+952.38	−1000.00
Net cash flow	+12.38	0.00

The opportunity for arbitrage will force the price of the bond to rise until it is equal to $952.38.

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	Today($)	In One Year($)
Sell the bond	+960.00	−1000.00
Invest at the bank	−952.38	+1000.00
Net cash flow	+ 7.62	0.00

The opportunity for arbitrage will force the price of the bond to rise until it is equal to $952.38.

Separation Principle

Normal market → security transactions should be zero-NPV
(since there isn't any arbitrage opportunity…)

So trading with securities doesn't create or destroy value →
FINANCIAL TRANSACTIONS: help to adjust the timing and risk of the cash flows

Value is created by real investments (*eg. Developing new products*)

Separation principle: We can evaluate the NPV of an ***investment decision separately*** from the firm's decision regarding ***how to finance*** the investment or any other security transactions the firm is considering.

Risky cash flows

1. When cash flows are risky, we must discount them at a rate equal to the risk-free interest rate plus an appropriate risk premium.

2. The appropriate risk premium will be higher the more the project’s returns tend to vary with overall risk in the economy.

The price of risk

Risk aversion: investors prefer to have a safe income rather than a risky one of the same average amount.

The personal cost of losing a dollar in bad times is greater than the benefit of an extra dollar in good times.
(1100$ → 1400$ is not as good as $1100 → $800 bad is)

Risky Versus Risk-free Cash Flows

Assume there is an equal probability of either a weak economy or strong economy.

		Cash Flow in One Year	
Security	**Market price Today**	**Weak Economy**	**Strong Economy**
Risk-free bond	1058	1100	1100
Market index	1000	800	1400

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Risk free return: 1100 / 1058 –1 = 4%

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Market index expected value:

$$\frac{1}{2}(\$800) + \frac{1}{2}(\$1400) = \$1100$$

Expected return:

1100 / 1000 -1 = 10%

Risk free return: 1100 / 1058 –1 = 4%

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Risk free return:

1100 / 1058 –1 = 4%

Difference is: (Market) Risk Premium → MRP → 6%

What is the price of security A?

	Cash Flow in One Year		
Security	**Market Price Today**	**Weak Economy**	**Strong Economy**
Risk- free bound	769	800	800
Security A	?	0	600
Market index	1000	800	1400

What is the price of security A?

		Cash Flow in One Year	
Security	Market Price Today	Weak Economy	Strong Economy
Risk- free bound	769	800	800
Security A	?	0	600
Market index	1000	800	1400

If we combine security A with a risk-free bond that pays \$800 in one year, the cash flows of the portfolio in one year are identical to the cash flows of the market index.

By the Law of One Price, the total market value of the bond and security A must equal \$1000, the value of the market index.

Price of A = \$1000 – \$769 = \$231

Expected return of A = (½ x 0 + ½ x 600) / 231 – 1 = 30% → risk premium is 26%. (r_f = 4% = 800 / 769 – 1)

Arbitrage with transaction cost

When there are transactions costs, arbitrage keeps prices of equivalent goods and securities close to each other. Prices can deviate, but not by more than the transactions cost of the arbitrage.

Problem

Consider a bond that pays $1000 at the end of the year. Suppose the market interest rate for deposits is 6%, but the market interest rate for borrowing is 6.5%. What is the no-arbitrage price **range** for the bond? That is, what is the highest and lowest price the bond could trade for without creating an arbitrage opportunity?

Problem solution

The no-arbitrage price for the bond equals the present value of the cash flows. In this case, however, the interest rate we should use depends on whether we are borrowing or lending. For example, the amount we would need to put in the bank today to receive $1000 in one year is:

$$(\$1000 \text{ in one year}) \div (1.06 \text{ \$ in one year/\$ today}) = \$943.40 \text{ today}$$

where we have used the 6% interest rate that we will earn on our deposit. The amount that we can borrow today if we plan to repay $1000 in one year is:

$$(\$1000 \text{ in one year}) \div (1.065 \text{ \$ in one year/\$ today}) = \$938.97 \text{ today}$$

where we have used the higher 6.5% rate that we will have to pay if we borrow.

Suppose the bond price P exceeded $943.40. Then you could profit by selling the bond at its current price and investing $943.40 of the proceeds at the 6% interest rate. You would still receive $1000 at the end of the year, but you would get to keep the difference $\$(P - 943.40)$ today. This arbitrage opportunity will keep the price of the bond from going higher than $943.40.

Alternatively, suppose the bond price P were less than $938.97. Then you could borrow $938.97 at 6.5% and use P of it to buy the bond. This would leave you with $\$(938.97 - P)$.